

Text Mining Final Mini Project Report

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Course	CS 411: Text Mining
Assignment	Final Mini Project Report

1. Introduction

Sentiment analysis is a Natural Language Processing (NLP) task that automatically classifies the emotional tone of a piece of text, like expressing a positive or negative opinion. This project applies classical NLP and machine learning techniques to the IMDB Movie Reviews dataset (50,000 reviews) to build, evaluate, and compare three binary sentiment classifiers.

Project Scope

- Preprocess raw text reviews into a clean, machine-readable format
- Extract numerical features using TF-IDF vectorization
- Train and compare three ML classifiers: Naive Bayes, Logistic Regression, and SVM
- Evaluate model performance using standard NLP metrics
- Perform error analysis and propose improvements

Dataset source: [Kaggle — IMDB Dataset of 50,000 Movie Reviews](#).

The dataset contains 50,000 reviews with equal representation of positive and negative sentiment (25,000 each), making it well balanced for binary classification.

2. Dataset Preparation & Exploratory Data Analysis

The dataset was loaded from IMDB Dataset.csv using pandas. Initial exploration confirmed the following:

Property	Value
Total reviews	50,000

Positive reviews	25,000 (50%)
Negative reviews	25,000 (50%)
Missing values	None
Columns	review (text), sentiment (positive/negative)

A binary label column was created by mapping positive $\rightarrow$ 1 and negative $\rightarrow$ 0 to prepare the data for scikit-learn classifiers. A histogram visualization confirmed the equal distribution visually.

3. Text Preprocessing

Raw movie reviews contain HTML tags, punctuation, numbers, and high-frequency common words that carry little predictive signal for sentiment. A custom preprocess_text() function was applied to all 50,000 reviews in the following order:

Step	Method	Purpose
1. Lowercase	str.lower()	Normalizes case so 'Great' and 'great' are treated identically
2. Remove HTML tags	re.sub(r'<[>]+'')	Strips and other HTML markup common in IMDB reviews
3. Remove non-alpha chars	re.sub(r'[^\w\s]')	Removes punctuation, numbers, and symbols
4. Collapse whitespace	re.sub(r'\s+', ' ') + strip()	Removes extra spaces introduced by previous steps
5. Tokenization	str.split()	Splits cleaned text into individual word tokens
6. Stopword removal	NLTK stopwords	Drops high-frequency words (e.g. 'the', 'is') with no sentiment value
7. Lemmatization	NLTK WordNetLemmatizer	Reduces words to dictionary base form (e.g. 'running' $\rightarrow$ 'run')

Lemmatization was chosen over stemming because it produces real dictionary words, which map better to TF-IDF vocabulary entries. For example, 'better' → 'good' and 'running' → 'run', whereas a stemmer might produce truncated non-words.

4. Feature Extraction — TF-IDF Vectorization

The cleaned reviews were converted to numerical features using TF-IDF (Term Frequency–Inverse Document Frequency) vectorization. The dataset was first split 70% training / 30% testing with stratified sampling to maintain class balance. The vectorizer was fit only on training data and then applied to test data to prevent data leakage. The following TF-IDF parameters are used in the process.

- `max_features = 5,000` — retains only the top 5,000 most informative tokens, reducing dimensionality
- `ngram_range = (1, 2)` — captures both single words (unigrams) and two-word phrases (bigrams) such as 'not good' or 'highly recommend'

After vectorization, each review is represented as a sparse matrix with 5,000 numerical features. The training set contains 35,000 samples and the test set contains 15,000 samples. Bigrams are essential for capturing negations like 'not great' that unigrams alone would misrepresent.

5. Model Training

Three classifiers were trained on the TF-IDF feature matrix. Each was chosen for its established effectiveness in text classification tasks:

Model	Key Parameters
Logistic Regression	<code>C = 0.1</code> , <code>max_iter = 1000</code>
Naive Bayes (MultinomialNB)	<code>alpha = 0.1</code>
SVM (LinearSVC)	<code>C = 0.1</code>

6. Model Evaluation & Results

6.1 Performance Comparison

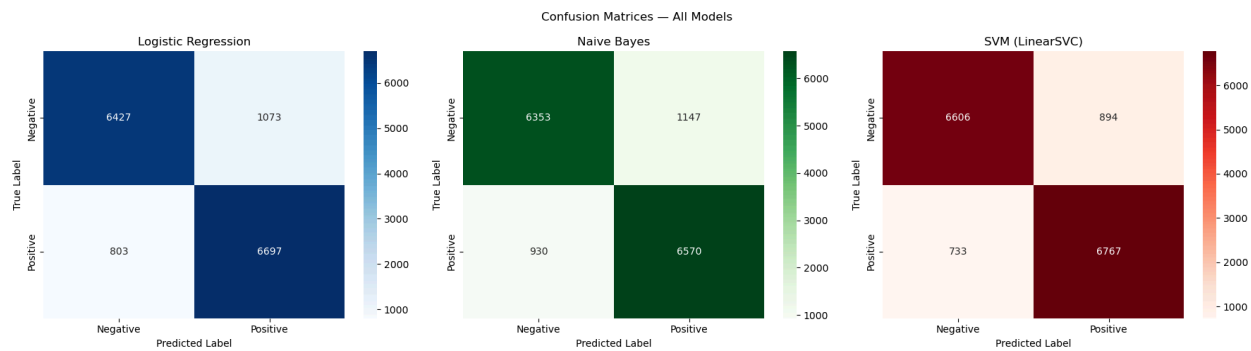
Each model was evaluated on the 15,000-sample test set using four metrics:

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.87	0.87	0.87	0.87
Naive Bayes (MultinomialNB)	0.86	0.86	0.86	0.86
SVM (LinearSVC)	0.89	0.89	0.89	0.89

SVM achieved the highest scores across all four metrics, consistent with its known strength in high-dimensional sparse text classification. Logistic Regression was a close second, making it a viable alternative when probability estimates are needed. Naive Bayes, despite its simplicity, achieved competitive results above 85% — impressive for a probabilistic model.

6.2 Classification Reports and Confusion Matrix Analysis

Confusion matrices were visualized as heatmaps (Blues for Naive Bayes, Greens for Logistic Regression, Reds for SVM).



- In all three heatmaps, the diagonal cells (True Negatives and True Positives) are the darkest, confirming the majority of predictions are correct
- The matrices are nearly symmetric, reflecting the balanced dataset
- SVM had the smallest off-diagonal values (False Positives and False Negatives), confirming its superiority

- False Positives and False Negatives were roughly equal across all models, indicating no systematic class bias

7. Error Analysis

Error analysis focused on the best-performing model (SVM). Misclassified reviews were extracted by comparing true labels against predicted labels, then analyzed for patterns. Four primary failure modes were identified as follows.

Failure Mode	Example	Root Cause
Sarcasm & Irony	"Oh brilliant, another two hours of my life I'll never get back."	TF-IDF treats words as independent features; cannot detect tone or context-dependent meaning
Short / Vague Reviews	"Not bad." / "Could have been better."	Short reviews produce sparse TF-IDF vectors with too few nonzero features to classify confidently
Mixed Sentiment	"Stunning cinematography, but the script was painfully weak."	Binary label forces one class; bag-of-words cannot represent multi-aspect sentiment
Domain Vocabulary	Rare film terms: 'mise-en-scene', 'hammy', 'saccharine'	max_features=5000 may exclude rare but informative tokens

All four cases require models with deeper language understanding to resolve effectively.

8. Suggested Improvements

#	Improvement	Expected Benefit
1	Use BERT / transformer models	Captures contextual word meaning, handles negation and sarcasm through attention mechanisms
2	Hyperparameter tuning with GridSearchCV	Systematically optimizes C, alpha, and max_features for best accuracy

3	Increase max_features (e.g. 20,000)	Retains more rare but informative vocabulary items (e.g. domain-specific film terms)
4	Add trigrams (ngram_range = (1,3))	Captures longer phrase-level sentiment patterns beyond bigrams
5	Combine with sentiment lexicons (VADER, SentiWordNet)	Adds external polarity signals for words the model may not have learned well

9. Conclusion

This project demonstrated that classical NLP pipelines combining TF-IDF feature extraction with linear machine learning models can achieve strong binary sentiment classification performance on large, balanced datasets.

Key findings:

- TF-IDF with unigrams and bigrams (max_features=5,000) provides an efficient and effective feature representation for sentiment classification
- SVM (LinearSVC) is the best-performing model due to its ability to find maximum-margin boundaries in high-dimensional sparse feature spaces
- Logistic Regression is a strong alternative when probability estimates are required, with only a marginal accuracy trade-off
- Naive Bayes, despite its simplicity, achieves above-85% accuracy, demonstrating viability for large balanced datasets
- Error analysis revealed that sarcasm, short reviews, mixed-sentiment, and domain vocabulary are the primary failure modes — all rooted in the context-free nature of bag-of-words representations

References

- Kaggle — IMDB Dataset of 50,000 Movie Reviews:
<https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews>